TrES-5b

R-0994

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-5_180825` · created 2026-07-30T00:14:50+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458355.8719 +/- 0.0025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.123 +/- 0.053
Transit depth (Rp/Rs)^2	1.51 +/- 1.3 [%]
Orbital Inclination (inc)	82.75 +/- 2.94
Airmass coefficient 1 (a1)	1.9526 +/- 0.00026
Airmass coefficient 2 (a2)	-0.41 +/- 0.19
Scatter in the residuals of the lightcurve fit is	0.009 %
Best Comparison Star	#2 - [206, 130]
Optimal Aperture	66.47
Optimal Annulus	98.67
Transit Duration (day)	0.0024 +/- 0.0037

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.123 ± 0.053 vs 0.142 (deviation 0.36σ)
Duration fit vs published	0.0024 d vs 0.0758 d (deviation 19.85σ)
Mid-time O–C	5.8 min
Depth SNR	2.3
Residual scatter	0.009 %

Light curve

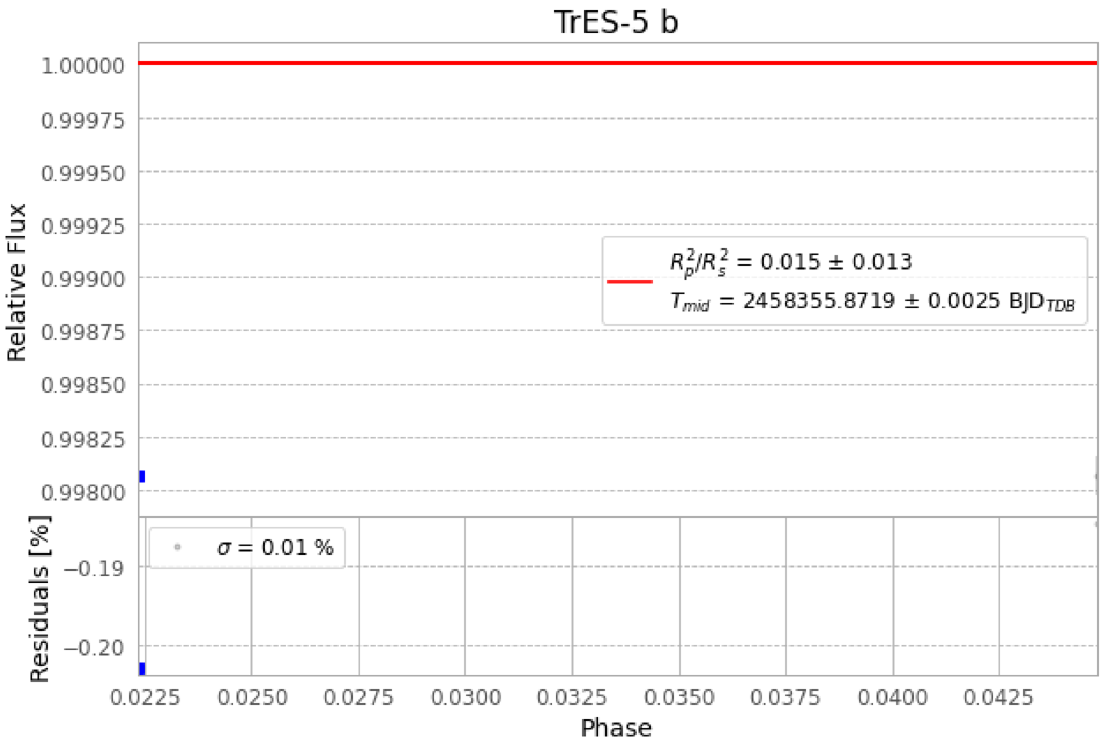

Figure 1 — FinalLightCurve_TrES-5 b_2018-08-24.png (SHA-256 9402f917e187a638…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [587, 241], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 073700a147fc35a0…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

77 FITS frames (51 MB), `TRES-5180825065115.FITS` → `TRES-5180825103912.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-180825.log (7aac1fa594d8…), FinalLightCurve_TrES-5 b_2018-08-24.png (9402f917e187…), FinalLightCurve_TrES-5 b_2018-08-24.csv (19d316682886…)

Compute resources

Wall time 746.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-30 00:14Z.